1,000 Labels Is All You Need: Non-Monotonic Scaling of Label Compliance in Fine-Tuned LLMs for Cybersecurity Classification

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***Abstract*—How many labeled examples does a fine-tuned LLM need to master a domain-specific classification task? We address this question by training Qwen3.5-9B with QLoRA on 1K, 5K, 10K, and 20K samples from the SALAD cybersecurity dataset and discover two distinct learning processes operating on different scales. *Semantic learning*—the ability to correctly classify alerts into attack categories—saturates at just 1K samples (normalized F1 = 100% at all sizes). *Label vocabulary learning*—the ability to express classifications using exact schema labels—follows a surprising non-monotonic curve: strict F1 drops from 87.5% at 1K to 83.6% at 5K before recovering to 97.4% at 10K and 100% at 20K. The 5K anomaly produces 4,925 label substitution errors (96× more than 1K's 51), as additional training capacity enables MITRE ATT&CK pre-training knowledge to override the task schema ("Reconnaissance" → "Port Scanning"). This phenomenon is stochastic: across 3 seeds at 5K, strict F1 ranges from 83.6% to 94.0% (mean 0.886 ± 0.053). LoRA hyperparameters modulate the effect: lower ranks (16, 32) achieve perfect compliance on clean data by limiting capacity for pre-training bleed-through. Reasoning-oriented models (DeepSeek-R1, Phi4-mini) achieve perfect strict F1 at 5K without data scaling, suggesting an architectural shortcut. We provide practical guidelines: 1K for semantic understanding ($0.50), 5K with reasoning models for full compliance ($3.72), or 20K for architecture-agnostic compliance ($4.48).**

***Index Terms*—sample efficiency, label hallucination, scaling laws, fine-tuning, cybersecurity NLP, QLoRA, MITRE ATT&CK**

I. Introduction

The conventional wisdom in machine learning suggests that more training data leads to monotonically better performance. Neural scaling laws [7] formalize this relationship for pre-training loss, demonstrating power-law improvement with increasing data, model size, and compute. Chinchilla [3] refines these relationships, showing that data should scale proportionally to model parameters for compute-optimal training. LIMA [8] extends these insights to fine-tuning, demonstrating that as few as 1,000 high-quality examples suffice for instruction tuning.

However, all these scaling laws measure a single dimension of performance: accuracy, loss, or semantic correctness. We identify a second dimension—*label compliance*—that follows fundamentally different scaling behavior. Label compliance measures whether a model uses the exact label vocabulary specified by the task schema, as opposed to semantically equivalent but lexically different alternatives.

We discover a surprising result: **semantic understanding requires only 1K samples, but label compliance follows a non-monotonic curve with a "knowledge activation valley" at 5K**. At 1K, the model correctly classifies alerts but makes 51 surface-level errors (e.g., "Backdoor" → "Backdoors"). At 5K, errors *increase* 96-fold to 4,925 as the model acquires sufficient capacity to express MITRE ATT&CK sub-technique names from its pre-training corpus. Only at 20K does the model learn to suppress pre-training vocabulary entirely.

This finding has immediate practical implications for organizations deploying fine-tuned LLMs in Security Operations Centers (SOCs), where downstream automation (SOAR playbooks, ticketing systems, compliance reporting) requires exact label matches. Our contributions:

1) We identify *semantic saturation* at 1K samples and *vocabulary compliance* at 20K, demonstrating that these are distinct learning processes.
2) We document the *knowledge activation valley*—a non-monotonic hallucination curve where intermediate training sizes (5K) produce more errors than smaller or larger sizes.
3) We characterize the interaction between LoRA hyperparameters (rank, learning rate), data quality (clean vs. noisy), and training size on label compliance.
4) We show that reasoning-oriented model architectures (DeepSeek-R1, Phi4-mini) bypass the scaling requirement entirely, achieving perfect compliance at 5K.
5) We provide cost-optimal training size guidelines for different compliance requirements.

II. Related Work

A. Neural Scaling Laws

Kaplan et al. [7] established power-law scaling for pre-training loss as functions of model size N, dataset size D, and compute budget C: $L(N) \propto N^{-\alpha_N}$, $L(D) \propto D^{-\alpha_D}$. Hoffmann et al. [3] (Chinchilla) showed that optimal training allocates compute equally between model size and data, leading to smaller but better-trained models. These laws describe *pre-training loss*, a holistic measure that does not capture label-level compliance in downstream tasks.

For fine-tuning specifically, Zhou et al. [8] (LIMA) demonstrated that 1K carefully curated examples suffice for instruction tuning of LLaMA-65B, achieving competitive performance with models trained on orders of magnitude more data. Our results agree with LIMA for *semantic understanding* but reveal that label compliance—essential for production deployment—requires 20× more data or architectural intervention. This distinction has not been previously documented.

B. LLM Hallucination in Classification

Hallucination in LLMs has been extensively studied for factual generation [6], [5]. Ji et al. [6] provided a comprehensive survey covering intrinsic and extrinsic hallucination in NLG tasks. Huang et al. [5] catalogued hallucination types across model architectures and proposed mitigation strategies.

Most relevant to our work, Gekhman et al. [2] showed that fine-tuning LLMs on new factual knowledge can paradoxically *increase* hallucination tendency—a finding directly consistent with our 5K anomaly. They demonstrated that models learn new knowledge slower than they learn to express familiar patterns, creating a window where hallucination increases. Our contribution extends this to classification tasks, where the "new knowledge" is the label vocabulary and the "familiar pattern" is the MITRE ATT&CK taxonomy from pre-training.

C. Parameter-Efficient Fine-Tuning

LoRA [4] decomposes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times d}$, with rank $r \ll \min(d, d)$ controlling the capacity of the update. QLoRA [1] combines LoRA with 4-bit NormalFloat quantization, enabling fine-tuning of 7B+ models on a single GPU with ¡16GB VRAM. We leverage QLoRA's efficiency to conduct a systematic scaling study across 4 training sizes, 4 ranks, 3 learning rates, and 2 data quality conditions—a 96-configuration grid that would be prohibitively expensive with full fine-tuning.

D. Data Quality and Leakage

Kapoor and Narayanan [**?**] documented widespread data leakage in ML evaluation, showing that leaked benchmarks overestimate model capabilities by 10–50%. Our dataset preparation addresses this explicitly: the original SALAD dataset contained 97.5% train-test overlap (993% leakage ratio by sample count). We demonstrate that data quality (clean vs. noisy/leaked) affects label compliance more than any hyperparameter choice, with the same rank 16 achieving 100% on clean data vs. 98.4% on noisy data.

III. Experimental Setup

A. Dataset and Data Integrity

We use the SALAD dataset (SOC Alert Labelled Annotated Dataset), derived from UNSW-NB15 [**?**]. Each sample contains 47 network flow features and 4 labels: Classification (2 classes, $H = 0.083$ bits), Triage Decision (3 classes, $H = 0.834$ bits), Attack Category (8 classes, $H = 1.240$ bits), and Priority Score (continuous, 0–1).

The original dataset contained severe data leakage. We created clean, deduplicated, stratified splits:

- 1K, 5K, 10K, 20K training subsets (nested, zero-overlap with test)
- 9,851 held-out test samples

All training subsets are stratified by attack category to maintain class distribution.

TABLE I
Training Configuration

Parameter	Value
Model	Qwen3.5-9B
Quantization	4-bit NF4 (QLoRA)
Compute dtype	bf16
LoRA rank (default)	64
LoRA alpha	128
Learning rate (default)	2×10^{-4}
LR scheduler	Cosine
Epochs	3
Batch size	8 (grad accum 4 = eff. 32)
Max seq length	2,048
Hardware	NVIDIA A100 80GB
Framework	LlamaFactory v0.9.2

B. Model and Training Configuration

C. Cross-Model Comparison

To contextualize the scaling behavior, we compare Qwen3.5-9B against 6 additional models at the 5K training size: Qwen3.5-0.8B (0.8B), SmolLM2-1.7B (1.7B), Phi4-mini (3.8B, reasoning), Mistral-7B-v0.3 (7B), DeepSeek-R1-Distill (7B, reasoning), and Qwen3-8B (8B).

D. Ablation Grid

We systematically vary two hyperparameters:

- **LoRA rank**: $r \in \{16, 32, 64, 128\}$
- **Learning rate**: $\eta \in \{1 \times 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}\}$

Each configuration is evaluated on both clean (zero-overlap) and noisy (original SALAD with leakage) data to isolate the effect of data quality.

E. Evaluation Metrics

- **Strict macro-F1**: Exact string match. "Port Scanning" $\neq$ "Reconnaissance."
- **Normalized macro-F1**: After alias mapping to canonical labels.
- **Semantic-compliance gap**: $\Delta_{\text{SC}} = F1_{\text{norm}} - F1_{\text{strict}}$.
- **Error count**: Total label substitution errors in test set.

IV. Results

A. Semantic Saturation at 1K

Normalized F1 = 100% at all training sizes (1K, 5K, 10K, 20K). From a semantic perspective, 1K samples from a low-entropy task ($H = 1.24$ bits) is sufficient to learn all 8 attack categories and 3 triage decisions. This confirms LIMA's finding [8] that small, high-quality datasets suffice for task understanding.

B. The Non-Monotonic Strict Scaling Curve

The strict F1 curve is non-monotonic: 5K (83.6%) is *worse* than 1K (87.5%), producing a 96× increase in errors (51 → 4,925). At 10K, errors decrease to 1,690, and at 20K they vanish entirely.

TABLE II
NON-MONOTONIC SCALING: STRICT F1 VS. TRAINING SIZE (QWEN3.5-9B, SEED 42)

Size	Strict F1	Norm F1	Errors	Primary Error
1K	0.875	1.000	51	Backdoor → Backdoors
5K	**0.836**	**1.000**	**4,925**	Recon → Port Scanning
10K	0.974	1.000	1,690	Recon → Port Scanning
20K	**1.000**	1.000	**0**	None

C. Error Type Evolution Across Scales

The nature of errors changes qualitatively across training sizes, revealing the underlying learning dynamics:

At 1K (51 errors): All errors are *morphological*—pluralization of "Backdoor" to "Backdoors." This is a shallow surface error indicating incomplete memorization of the exact label string. The model has learned the concept but not the precise token sequence.

At 5K (4,925 errors): A fundamentally different error type emerges. 4,895 of 4,925 errors (99.4%) are *taxonomy substitutions*: "Reconnaissance" → "Port Scanning." The model has now activated its MITRE ATT&CK pre-training knowledge and is expressing the more specific sub-technique name. The remaining 30 errors are morphological ("Backdoors").

At 10K (1,690 errors): The same taxonomy substitution persists but at reduced frequency (1,690 vs. 4,925). The model has begun learning the schema vocabulary but hasn't fully suppressed pre-training alternatives.

At 20K (0 errors): Complete label compliance. The model has seen sufficient repetitions of each schema label to override pre-training vocabulary entirely.

This evolution reveals three competing learning phases:

1) **Phase 1—Shallow learning (1K):** Feature-to-category mappings with minimal pre-training activation. Errors are surface-level.
2) **Phase 2—Knowledge activation (5K):** Additional training unlocks MITRE ATT&CK pre-training pathways. The model "knows too much" and expresses it.
3) **Phase 3—Vocabulary override (10K–20K):** Sufficient schema label repetition teaches suppression of pre-training alternatives.

D. Seed Sensitivity at the Valley

TABLE III
SEED SENSITIVITY AT 5K: THE STOCHASTIC VALLEY

Seed	Strict F1	Errors	Primary Hallucination
42	0.836	4,925	Recon → Port Scanning
123	0.940	33	Backdoor → Backdoors
2024	0.881	83	Mixed sub-techniques
Mean ± std	0.886 ± 0.053		

The knowledge activation valley at 5K is highly sensitive to random initialization. Seed 42 triggers massive taxonomy substitution (4,925 errors), while Seed 123 largely avoids it (33 errors). The 10.4-point range across 3 seeds on identical data, model, and hyperparameters demonstrates that Phase 2 activation is *stochastic*: different initial conditions lead to different pre-training pathways being activated or suppressed.

This has critical implications for reproducibility: a single-seed evaluation at 5K could report anywhere from 83.6% to 94.0% strict F1, leading to drastically different conclusions about model quality. We recommend $\geq$3 seeds for any evaluation at intermediate training sizes.

E. Cross-Model Comparison at 5K

TABLE IV
7-MODEL COMPARISON AT 5K TRAINING SAMPLES

Model	Size	Strict F1	Errors	Type
DeepSeek-R1	7B	**1.000**	0	Reasoning
Phi4-mini	3.8B	**1.000**	0	Reasoning
Qwen3.5-0.8B	0.8B	0.875	51	Base
SmolLM2-1.7B	1.7B	0.875	51	Base
Qwen3.5-9B	9B	0.836	4,925	Base
Qwen3-8B	8B	0.753	4,947	Base
Mistral-7B	7B	0.749	119	Base

The cross-model comparison reveals that model architecture—specifically, reasoning-oriented training—completely eliminates the scaling requirement for label compliance. DeepSeek-R1 and Phi4-mini achieve perfect strict F1 at 5K, while all base models exhibit some degree of label hallucination. Notably, model size does not predict compliance: the 0.8B model (87.5%) outperforms the 9B (83.6%) and 8B (75.3%) models.

F. LoRA Hyperparameter Effects

TABLE V
LORA RANK VS. STRICT F1 (CLEAN DATA, 5K, LR=2E-4)

Rank	Strict F1	Norm F1	Errors
16	**1.000**	1.000	0
32	**1.000**	1.000	0
64	0.836	1.000	4,925
128	0.983	1.000	51

1) Rank Ablation on Clean Data:

TABLE VI
LORA RANK VS. STRICT F1 (NOISY DATA, 5K, LR=2E-4)

Rank	Strict F1	Norm F1	Errors
16	0.984	1.000	51
32	0.794	1.000	1,103
64	0.557	1.000	5,891
128	0.935	1.000	421

2) Rank Ablation on Noisy Data:

3) Learning Rate Ablation:

Key findings from the ablation grid:

(1) Lower ranks achieve better compliance on clean data. Ranks 16 and 32 achieve perfect strict F1, while rank 64 triggers knowledge activation. Lower rank = fewer dimensions for pre-training vocabulary to surface.

TABLE VII
LEARNING RATE VS. STRICT F1 (CLEAN DATA, 5K, RANK=64)

Learning Rate	Strict F1	Errors
1×10^{-4}	**1.000**	0
2×10^{-4}	0.836	4,925
5×10^{-4}	0.754	6,102

(2) Data quality dominates rank selection. The same rank 16 achieves 100% on clean data vs. 98.4% on noisy data. Rank 64 drops from 83.6% (clean) to 55.7% (noisy)—a 27.9-point degradation.

(3) Lower learning rates improve compliance. LR 1×10^{-4} achieves perfect compliance at rank 64 (clean data), recovering the same performance as lower ranks. Higher LRs cause larger gradient steps that escape the compliance-preserving loss basin.

(4) Rank 128 shows non-monotonic behavior. On noisy data, rank 128 (93.5%) outperforms both rank 32 (79.4%) and rank 64 (55.7%), suggesting very high capacity can also help override pre-training—the same U-shaped pattern as training size scaling.

V. DISCUSSION

A. A Two-Process Model of Fine-Tuning

Our results suggest that LLM fine-tuning involves (at least) two distinct learning processes:

1) **Semantic learning**: Mapping input features to output categories. This process is fast, saturating at ∼1K samples for low-entropy tasks ($H = 1.24$ bits). It leverages pre-training knowledge effectively.
2) **Vocabulary learning**: Mapping output categories to specific label strings. This process is slow, requiring 20K samples to override pre-training vocabulary. It *competes* with pre-training knowledge rather than leveraging it.

The non-monotonic curve at 5K arises from the interaction between these processes: Process 1 (semantic learning) activates pre-training knowledge (beneficial), but Process 2 (vocabulary learning) hasn't yet accumulated enough evidence to constrain the vocabulary (harmful). The net effect is more hallucination at 5K than at 1K.

B. Relationship to Gekhman et al. (2024)

Gekhman et al. [2] showed that fine-tuning on *new* knowledge increases hallucination of *existing* knowledge. Our finding is the converse: fine-tuning on schema labels (new vocabulary) is overwhelmed by pre-training vocabulary (existing knowledge) at intermediate training sizes. Both phenomena reflect the same underlying tension between new and existing knowledge, operating in different directions.

C. Comparison with LIMA

Zhou et al. [8] showed 1K examples suffice for instruction tuning of a 65B model. Our results agree for semantic understanding but add a critical nuance: **label compliance requires 20× more data** (20K vs. 1K). For applications where exact label strings matter—SOC automation, medical coding, legal classification—the LIMA finding significantly underestimates data requirements.

D. The Architecture Shortcut

Reasoning models (DeepSeek-R1, Phi4-mini) achieve perfect compliance at 5K without the knowledge activation valley. We hypothesize two mechanisms:

1) **Format compliance training**: Instruction-following and chain-of-thought training teaches precise output format adherence.
2) **Decoupled generation**: Reasoning models learn to separate internal reasoning from output generation, reducing vocabulary bleed-through.

E. Practical Training Size Guidelines

TABLE VIII
DECISION GUIDE FOR TRAINING SIZE SELECTION

Strategy	Data	Strict F1	Cost
Semantic only + post-proc	1K	87.5%	\$0.50
Avoid: base models	5K	83.6%	\$1.87
+ reasoning model	5K	100%	\$3.72
+ more data (any model)	10K	97.4%	\$2.24
Architecture-agnostic	20K	100%	\$4.48
+ low rank (16) + low LR	5K	100%	\$1.87

The most cost-effective path to 100% compliance is using a reasoning model at 5K (\$3.72) or tuning LoRA hyperparameters (rank 16, LR 1×10^{-4}) at 5K (\$1.87). **Training at 5K with default hyperparameters and a base model is the worst-case scenario.**

F. Generalization and Entropy Dependence

Our results are obtained on a low-entropy task ($H = 1.24$ bits, 8 classes). For higher-entropy tasks, we predict:

- **Semantic saturation**: Shifts rightward (more data needed). For GoEmotions ($H = 3.75$) and LedGAR ($H = 6.16$), we estimate 5K–20K for saturation.
- **Knowledge activation valley**: Shifts rightward and deepens. More classes means more opportunities for pre-training vocabulary to interfere.
- **Vocabulary compliance**: May require >20K for high-entropy tasks. We hypothesize scaling as $N_{\text{comply}} \propto 2^H \times k$ where k depends on overlap between task labels and pre-training vocabulary.

G. Threats to Validity

Internal: Single base model (Qwen3.5-9B); the valley may not occur in all architectures. Only 3 seeds tested; more seeds would tighten confidence intervals.

External: SALAD is a low-entropy cybersecurity task. Scaling curves may differ qualitatively for NLI, sentiment, or multilabel tasks.

Construct: Our strict F1 metric is binary—all label mismatches are equal. A graduated metric (sub-technique vs. cross-category) could provide finer resolution.

VI. Conclusion

Sample efficiency in LLM fine-tuning is not a single curve but two distinct processes: semantic learning saturates at 1K while vocabulary compliance requires up to 20K. The non-monotonic "knowledge activation valley" at 5K—where errors increase 96× relative to 1K—reveals a fundamental competition between task-specific learning and pre-training knowledge expression.

Practitioners should select training size based on compliance requirements: 1K for semantic-only applications with post-processing ($0.50), 5K with reasoning models or tuned LoRA for full compliance ($1.87–$3.72), or 20K for architecture-agnostic compliance ($4.48). Training at 5K with default hyperparameters and base models is the *worst-case* scenario—a counter-intuitive finding that challenges the "more data is always better" assumption.

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